

Overview

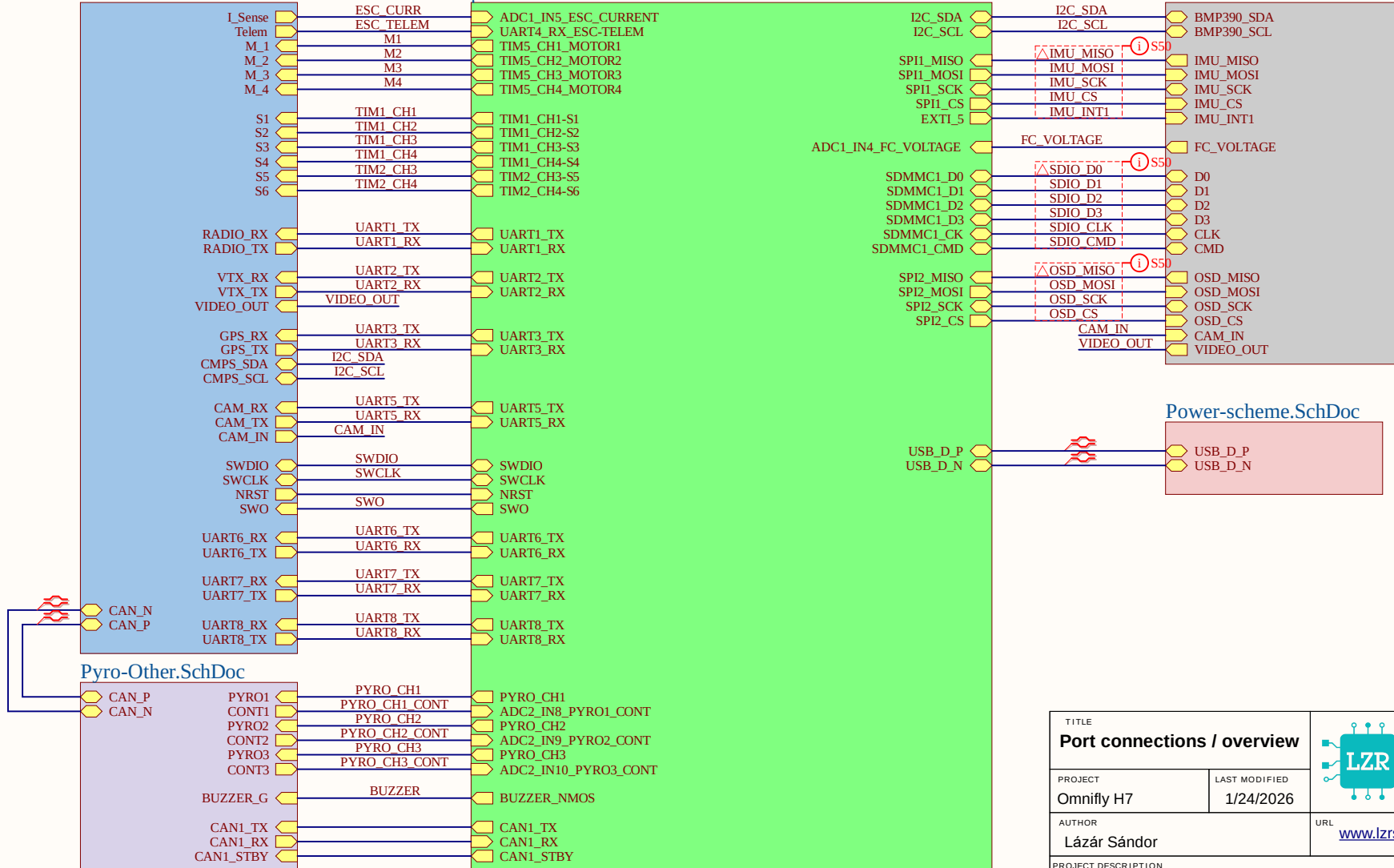
Headers-SolderPads.SchDoc

µC-Core.SchDoc

Sensors.SchDoc

Power-scheme.SchDoc

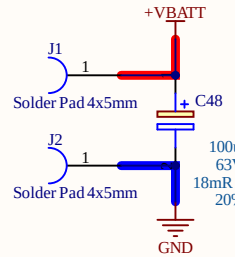
Pyro-Other.SchDoc



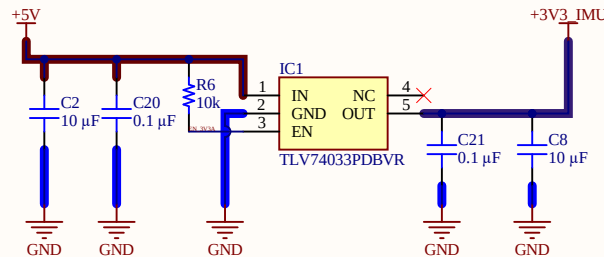
TITLE Port connections / overview			
PROJECT Omnifly H7	LAST MODIFIED 1/24/2026		
AUTHOR Lázár Sándor			
PROJECT DESCRIPTION Multi Platform Flight controller			REVISION 1
FILE NAME Overview.SchDoc		SHEET 1 OF 6	

Power Scheme

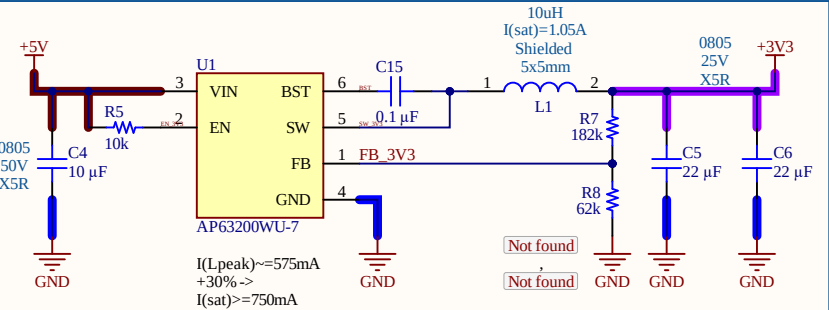
Direct battery voltage(2-6S)



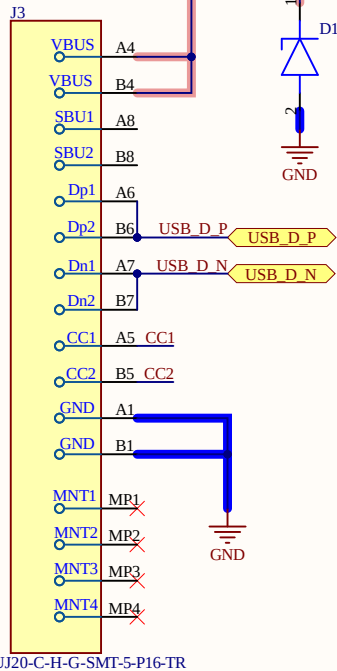
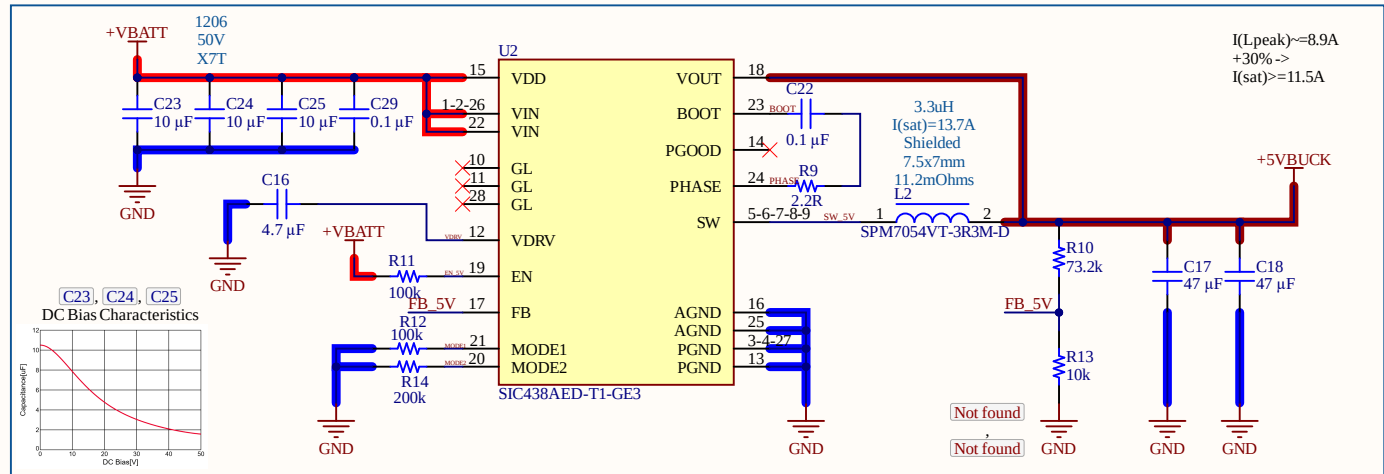
IMU LDO Regulator



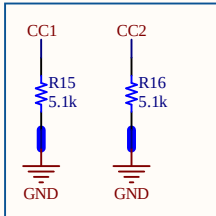
3.3V Buck Converter



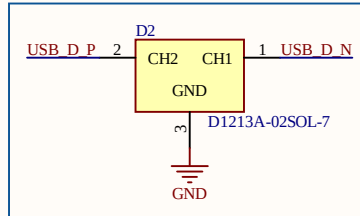
5V Buck Converter



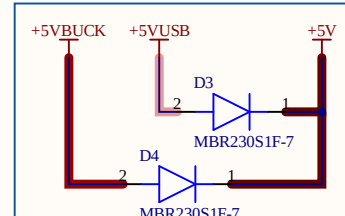
CC1, CC2 Pulldowns



USB D_P, D_N ESD Protection

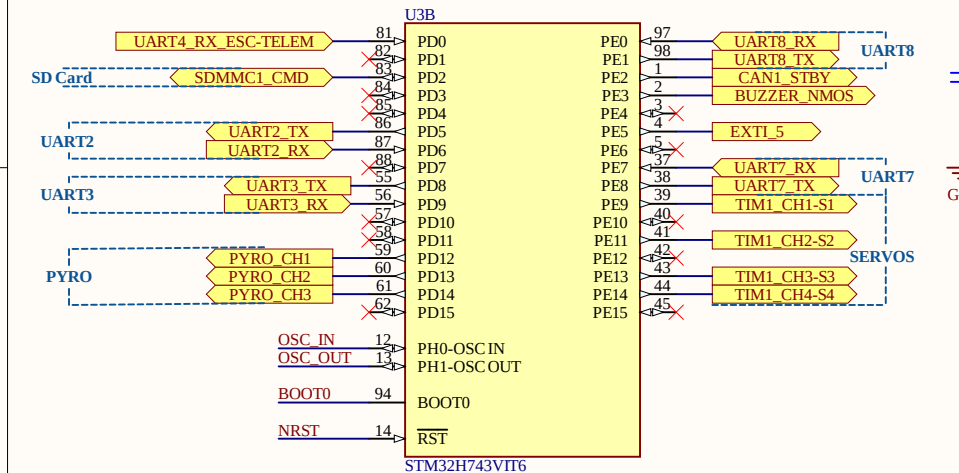


5V Rail



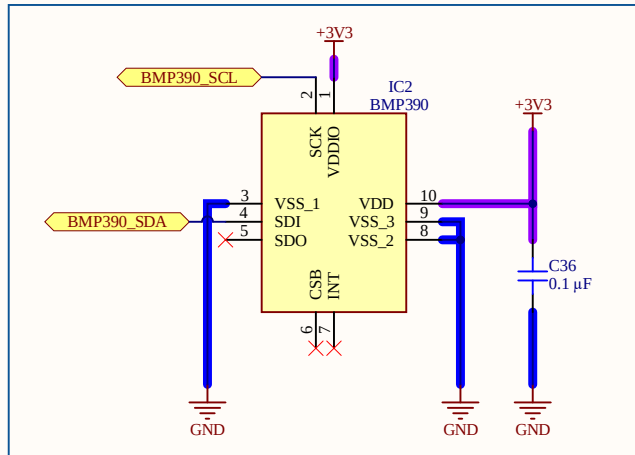
TITLE Power Scheme		LZR Systems	
PROJECT OmniFly H7	LAST MODIFIED 1/24/2026	URL www.lzrsystems.com	
AUTHOR Lázár Sándor		REVISION 1	
PROJECT DESCRIPTION Multi Platform Flight controller			1
FILE NAME Power-scheme.SchDoc			SHEET 2 OF 6

Microcontroller

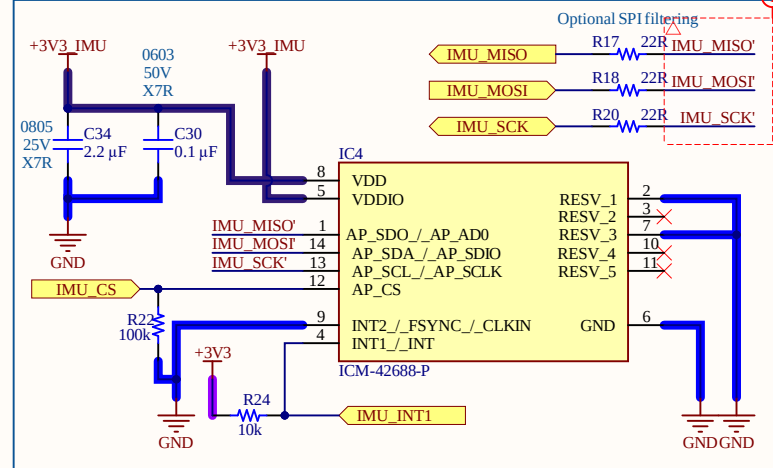


Sensors

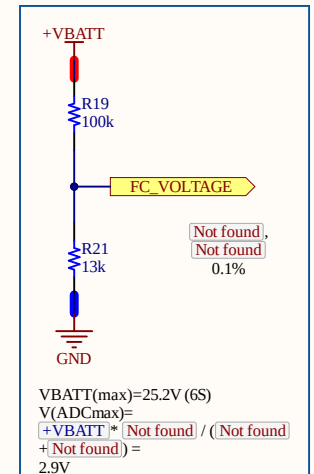
BMP390 - Barometric pressure sensor



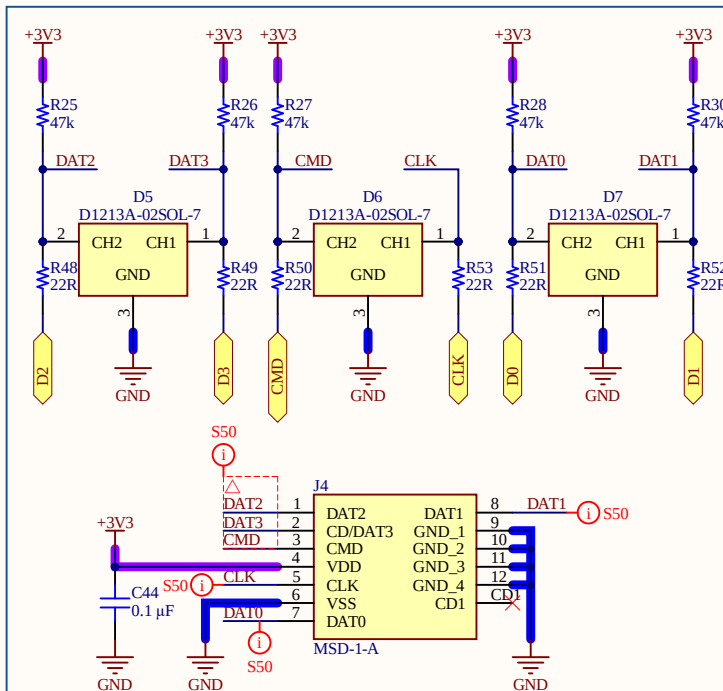
ICM-42688-P - Inertial measurement unit



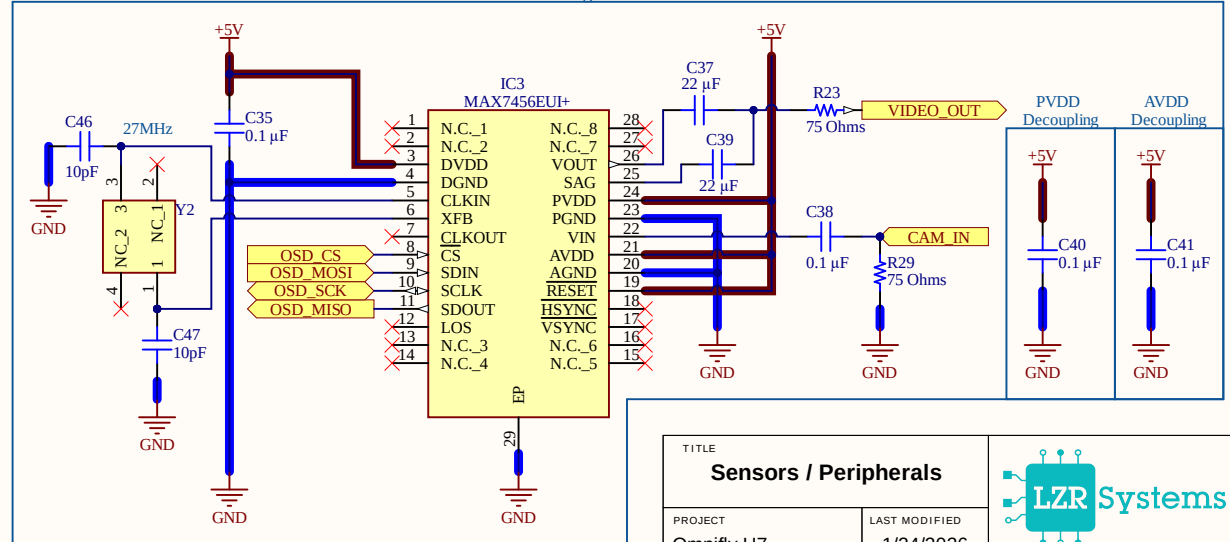
Voltage sense



MSD-1-A - micro SD Card reader



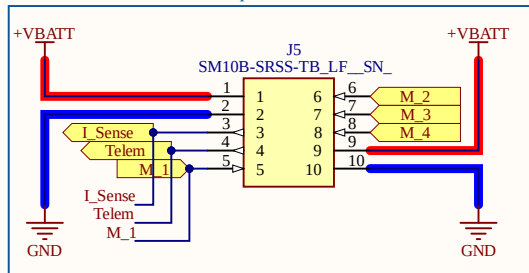
Analog video OSD interface



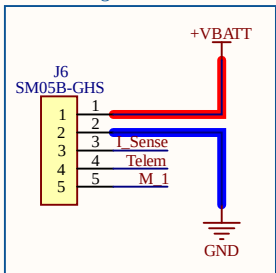
TITLE		Sensors / Peripherals		 LZR Systems	
PROJECT		LAST MODIFIED		www.lzrsystems.com	
OmniFly H7		1/24/2026			
AUTHOR			URL		
Lázár Sándor			www.lzrsystems.com		
PROJECT DESCRIPTION				REVISION	
Multi Platform Flight controller				1	
FILE NAME			SHEET 4 OF 6		
Sensors.SchDoc					

Headers-SolderPads

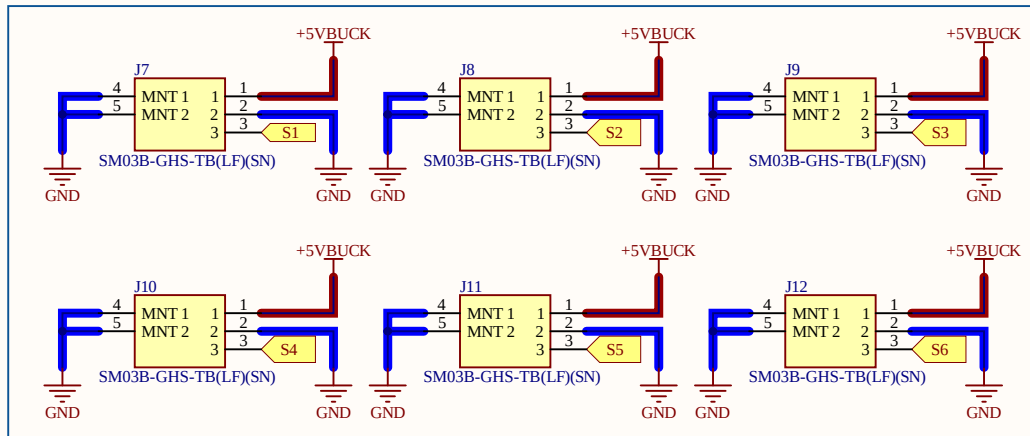
Standard quad motor ESC header



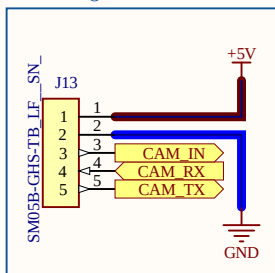
Custom single motor ESC header



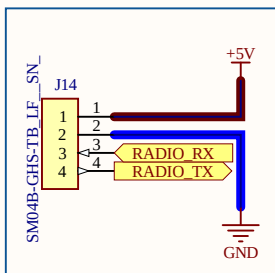
Servo headers



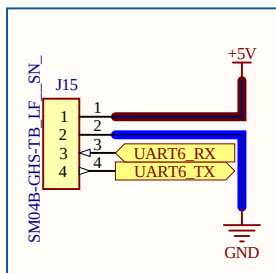
Analog FPV Camera header



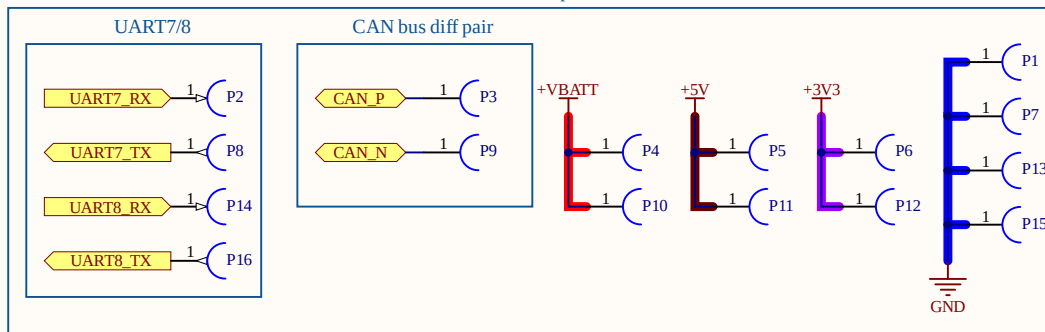
Radio / Remote header



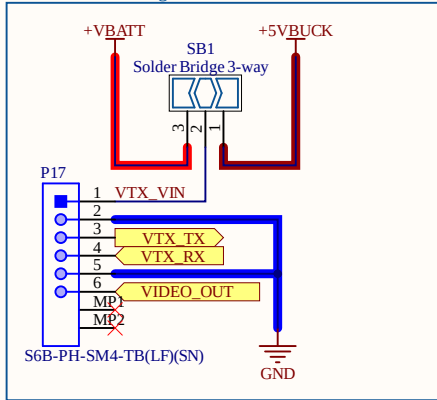
UART6 header



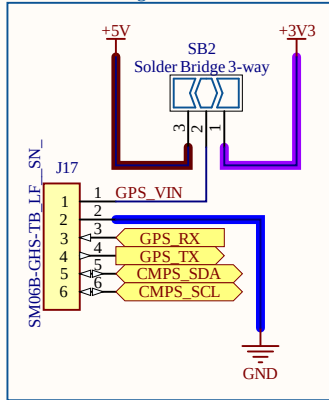
Solder pads



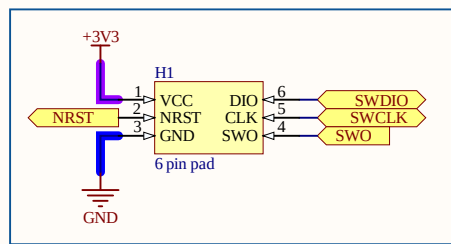
Analog VTX / DJI O4 header



GPS+Magnetometer header

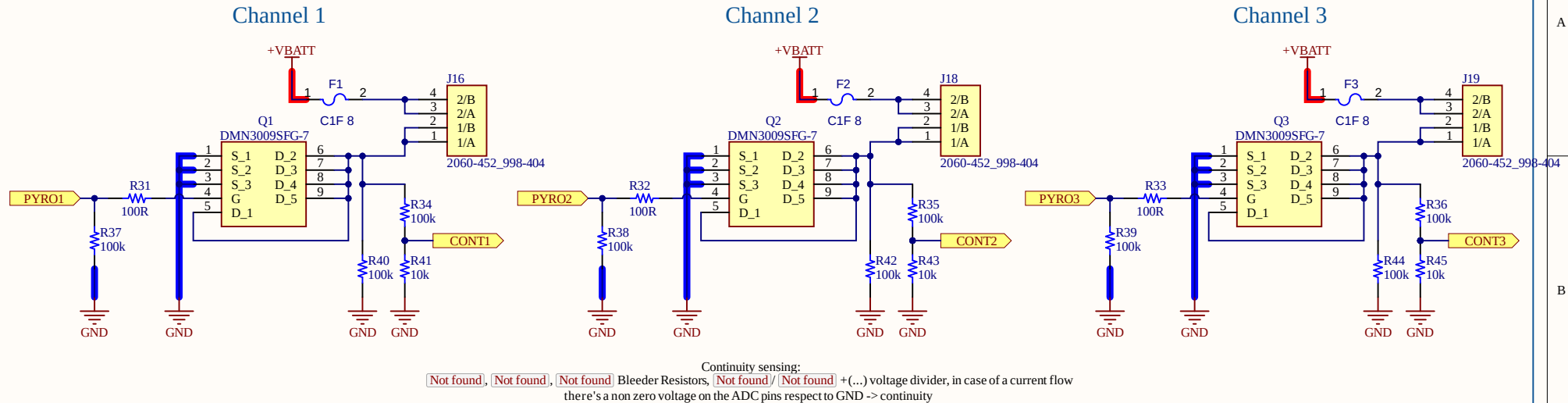


SWD header

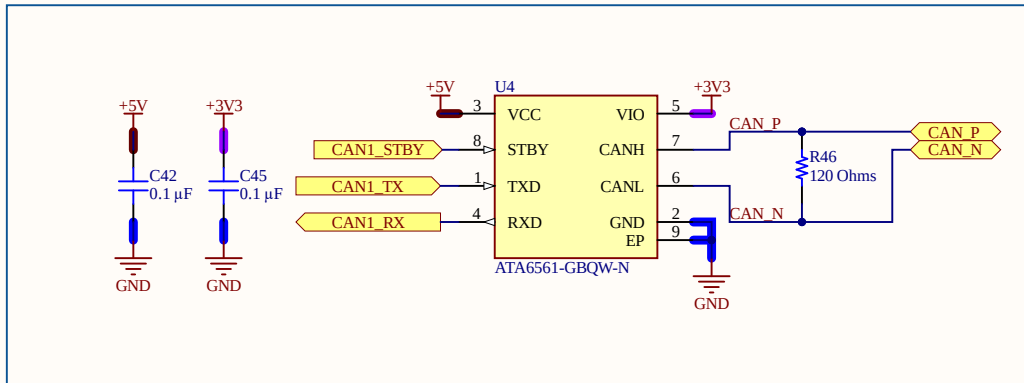


TITLE Headers / Solder pads			
PROJECT Omnifly H7	LAST MODIFIED 1/23/2026		
AUTHOR Lázár Sándor		URL www.lzrsystems.com	
PROJECT DESCRIPTION Multi Platform Flight controller			REVISION 1
FILE NAME Headers-SolderPads.SchDoc		SHEET 5 OF 6	

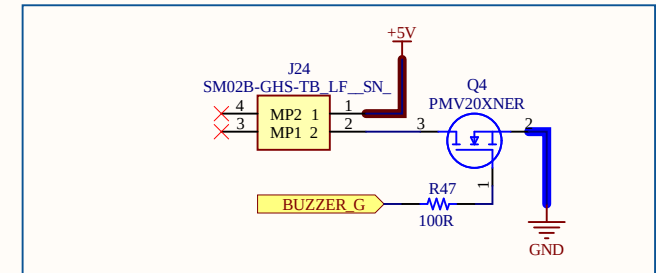
Pyro / Other peripherals



CAN bus transceiver



Buzzer



TITLE Pyro / Buzzer / CAN			
PROJECT Omnifly H7	LAST MODIFIED 1/24/2026		
AUTHOR Lázár Sándor		URL www.lzrsystems.com	
PROJECT DESCRIPTION Multi Platform Flight controller			REVISION 1
FILE NAME Pyro-Other.SchDoc			SHEET 6 OF 6

